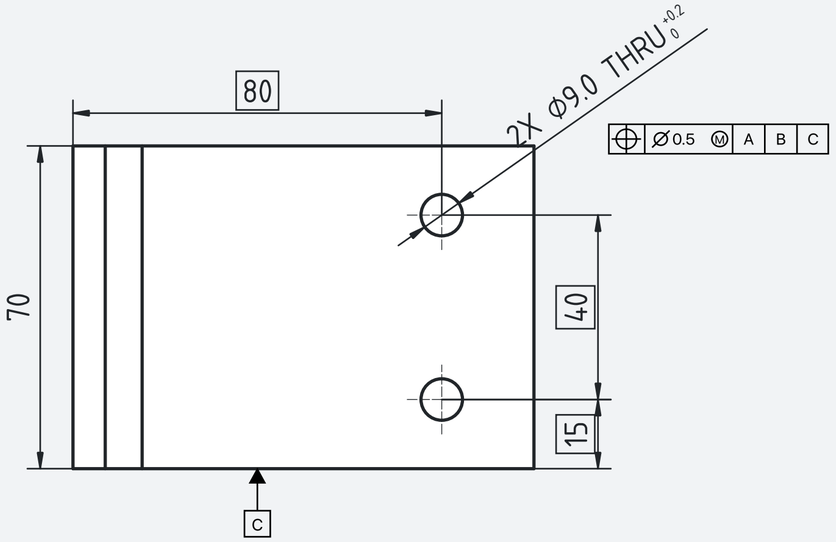
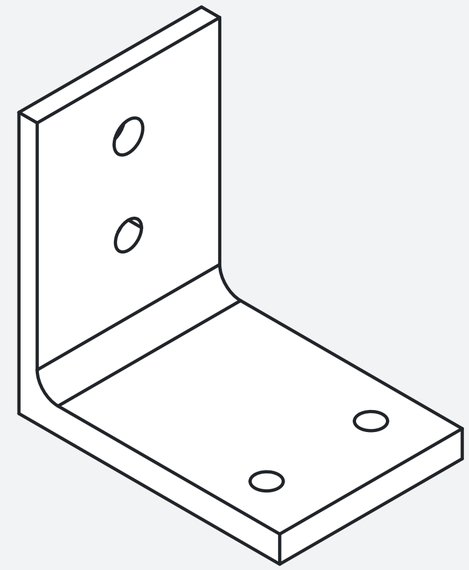
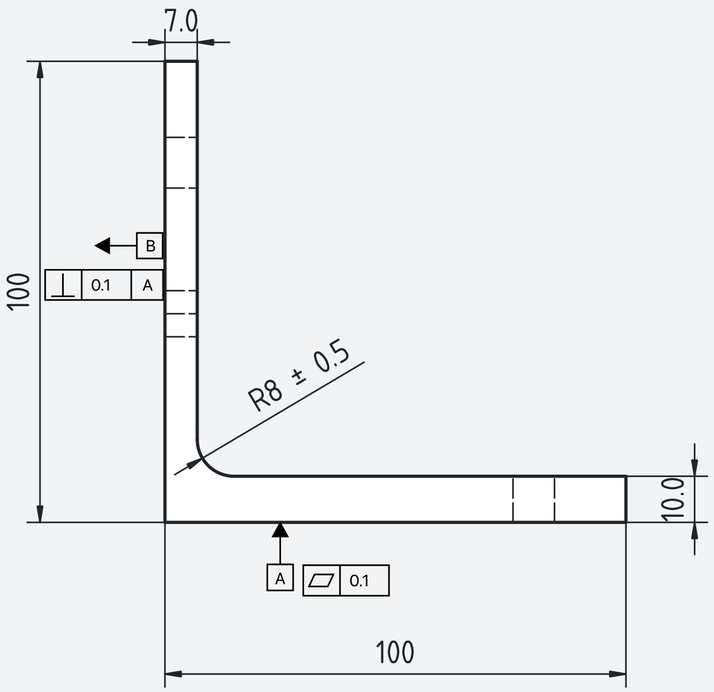
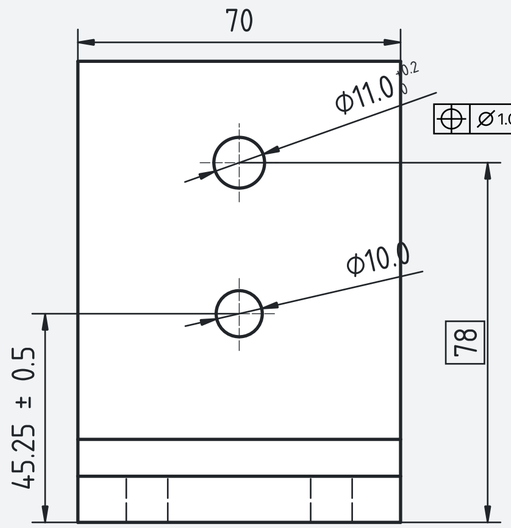








NOTES (UNLESS OTHERWISE SPECIFIED):

1. DIMENSIONS IN MILLIMETERS. TOLERANCES: X ±0.5 X.X ±0.2 X.XX ±0.1
2. INTERPRET DIMENSIONS AND TOLERANCES PER ASME Y14.5-2018.
3. MATERIAL: AISI 1020 HOT-ROLLED STEEL (Sy 210 MPa, Sut 380 MPa).
4. FILLET R8 IS FATIGUE-CRITICAL: Ra 1.6 MAX, NO TOOL MARKS ACROSS THE FILLET.
5. BREAK SHARP EDGES 0.2–0.5 EXCEPT FILLET R8.
6. FINISH: BLACK OXIDE.
7. BOXED DIMENSIONS ARE BASIC (THEORETICALLY EXACT).
8. MASS 0.884 kg (REF). DESIGN VERIFIED BY FEA, SEE bracket-fea/cad/README.

DESIGN VERIFICATION (CalculiX FEA OF THIS CAD GEOMETRY, FINE MESH):
FIRST NATURAL FREQUENCY: 107.8 Hz (REQUIRED: MIN 100 Hz)
PEAK STATIC STRESS AT 5/5/10 g: 16.9 MPa (ALLOWED: MAX 140 MPa)
RANDOM VIBRATION FATIGUE LIFE: 2312 h (REQUIRED: MIN 1000 h)
LOADS AND REQUIREMENTS ARE PROJECT ASSUMPTIONS.

DESIGNED BY: H. HO	**L-BRACKET**		I A RELEASE
DATE: 2026-11-04	COMPONENT MOUNT, MACHINED		H
CHECKED BY: -			G
DATE: -			F
SIZE ANSI C	PORTFOLIO PROJECT (NOT FOR PRODUCTION)		E
			D
SCALE 1:1	WEIGHT (kg) 0.884	DRAWING NUMBER BRK-001 REV A	SHEET 1 / 2	C
			B
github.com/RonHooo			A